

Procurement Bill of Materials

Module-level ordering guide (no vendor links)

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Platform: esp32

Total line items: 11

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Procurement note: Verify connector orientation, voltage levels, and form factor before ordering.

#	Item	Qty	Exactness	Ref	Required specs
1	ESP32 DevKit v1	1	Compatible	esp32	This project targets the ESP32 DevKit v1 (Espressif · esp32). • Match USB connector style and pinout.
2	NBL-S-TMC-7 7-in-1 Soil Composite Sensor	1	Compatible	TMC-7	The NBL-S-TMC-7 is a 7-in-1 soil composite sensor that integrates multiple soil parameters including temperature, moisture/humidity, electrical conductivity (EC), nitrogen (N), phosphorus (P), potassium (K), and pH into a single probe. It measures soil dielectric constant to directly and stably reflect real moisture content across various soil types. The probe is made of stainless steel for corrosion resistance and stable performance. The input power supply, sensing probe, and signal output sections are fully isolated for safety and reliability. Communication is via RS485 (Modbus RTU protocol). Suitable for soil moisture monitoring, scientific experiments, water-saving irrigation, greenhouses, flowers, vegetables, grass pastures, rapid soil testing, plant cultivation, precision agriculture, and related applications.
3	RS485 to TTL Module (MAX485)	1	Compatible	RS485	Interface: UART • Power: 5V • Module form factor varies by seller; match pin labels and connector layout.
4	DHT22	1	Compatible	DHT22	Power: 3.3V • Approx size: 15.1×20.5 mm
5	Rain Sensor Module	1	Generic	—	Module form factor varies by seller; match pin labels and connector layout.
6	YF-S201 Flow Sensor	1	Compatible	S201	YF-S201 is a Hall effect-based water flow measurement sensor with high-grade sealing. It outputs a pulse signal proportional to flow rate, with frequency (Hz) = 7.5 × flow rate (L/min). Operating voltage range is 4.5V to 18V DC, flow rate range is 1–30 L/min, and it produces approximately 450 pulses per liter with a 50% ±10% duty cycle.
7	12V DC Submersible Mini Water Pump	1	Generic	—	Two-wire brushed DC submersible water pump for small irrigation or circulation projects. It is an external load and must be driven through a relay, MOSFET, or motor driver; do not connect directly to an MCU GPIO.
8	12V Solenoid Valve	1	Generic	—	Power-path component: 12V Solenoid Valve. Sits on the power rail, not on a GPIO.
9	Relay Module	1	Generic	—	Power: 5V • Approx size: 50×26 mm • Module form factor varies by seller; match pin labels and connector layout.
10	Buck Converter	1	Generic	—	Power-path component: Buck Converter. Sits on the power rail, not on a GPIO.
11	12V DC Power Supply	1	Generic	—	Power-path component: 12V DC Power Supply. Sits on the power rail, not on a GPIO.